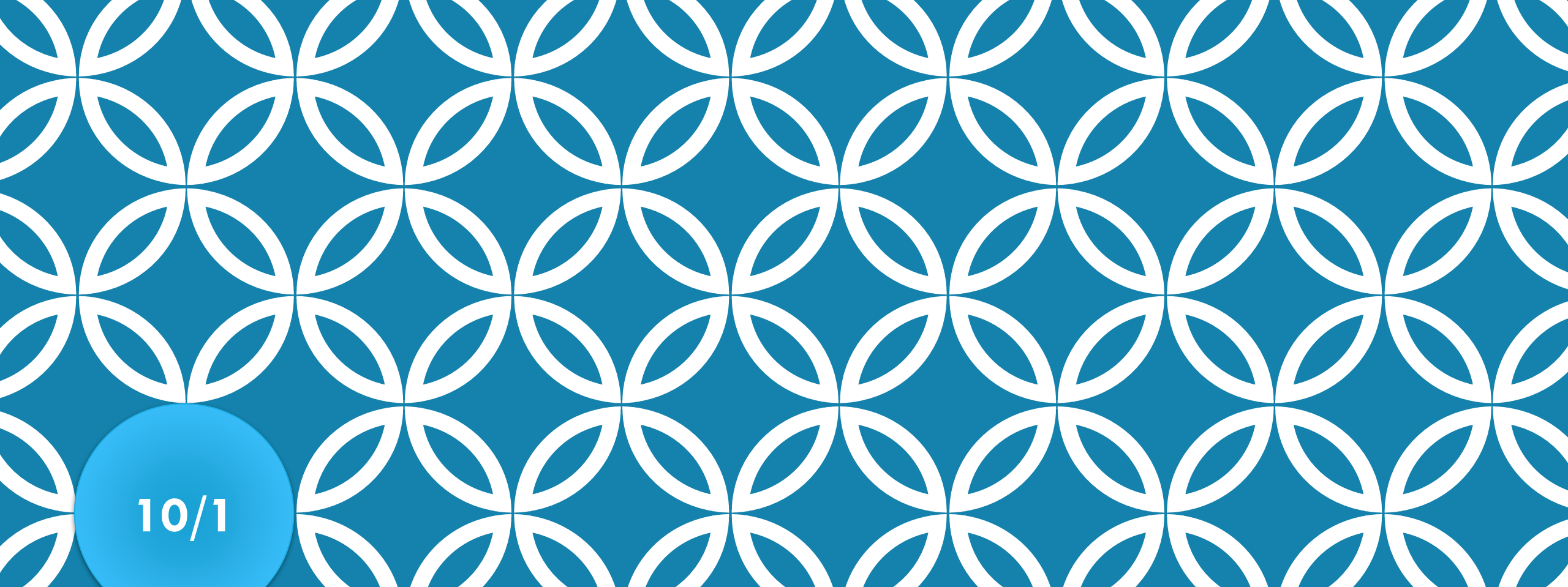




10/1

First-order logic

Bound and Free variables

Open and Closed formulas

Logic in Computer Science Lab
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Based on prof. Tamás Kádek's lectures

First-Order Language

An ordered 5-tuple $\mathcal{L}^{(1)} = \langle LC, Var, Con, Term, Form \rangle$ defines the language of the first-order (predicate) logic, where

1. $LC = \{ \neg, \supset, \wedge, \vee, \equiv, (,), =, \exists, \forall, , \}$ is the set of *logical constants*,
2. Var is the countable infinite set of variables,
3. $Con \neq \emptyset$; is the set of *non-logical constants* such that $LC \cap Con = \emptyset$, and

$$Con = \bigcup_{i=0}^{\infty} (\mathcal{P}(i) \cup \mathcal{F}(i));$$

4. $Term$ is the set of *terms*;
5. $Form$ is the set of *formulae*;

where LC , Var , $\mathcal{P}(i)$ and $\mathcal{F}(i)$ are pairwise disjoint.

The LC , Var and Con sets contains the letters of the alphabet, while the terms and formulas as sequences of letters are the words of the first-order language.

First-order language

Non-logical constants

The symbols of the Con set can be divided into two parts.

$$Con = \bigcup_{i=0}^{\infty} (\mathcal{P}(i) \cup \mathcal{F}(i));$$

1. symbols used as name of functions:

- $\mathcal{F}(0)$ contains the name parameters,
- $\mathcal{F}(k)$ where $k \geq 1$ contains the function parameters.

2. symbols used as name of predicates:

- $\mathcal{P}(0)$ contains the propositional parameters,
- $\mathcal{P}(k)$ where $k \geq 1$ contains the predicate parameters.

The argument k in $\mathcal{F}(k)$ and $\mathcal{P}(k)$ describes the arity (number of arguments) of the function and predicate parameters.

Logical constants

- the unary connective:
 - $\neg$ negation sign
- the binary connectives:
 - $\wedge$ conjunction sign
 - $\vee$ disjunction sign
 - $\supset$ implication sign
 - $\equiv$ sign of material equivalence
- the quantifiers:
 - $\exists$ existential quantifier
 - $\forall$ universal quantifier
- the equal sign: $=$
- the parentheses: $($ and $)$
- the separator sign: $,$

First-order terms

The set of $Term$ is defined inductively:

1. $Var \subseteq Term$;
2. $\mathcal{F}(0) \subseteq Term$;
3. if $f \in \mathcal{F}(k)$ for some $k \geq 1$ and $t_i \in Term$ for all $i = 1, \dots, k$, then

$$f(t_1, t_2, \dots, t_k) \in Term.$$

First-order formula

The set of *Form* is defined inductively:

Atomic formula

1. $\mathcal{P}(0) \subseteq \text{Form}$;
2. $(t_1 = t_2) \in \text{Form}$ if
 - t_1 and t_2 are terms ($t_1 \in \text{Term}$ and $t_2 \in \text{Term}$)
3. $f(t_1, t_2, \dots, t_k) \in \text{Form}$ if
 - $f \in \mathcal{P}(k)$ for some $k \geq 1$ and
 - $t_i \in \text{Term}$ for all $i = 1, \dots, k$.

First-order formula

Non atomic (compound) formula

4. $\neg A \in Form$ if A is a formula ($A \in Form$);
5. $(A \circ B) \in Form$ if
 - A and B are formulas ($A \in Form, B \in Form$) and
 - $\circ$ is a binary connective ($\circ \in \{\wedge, \vee, \supset, \equiv\}$);
6. $QxA \in Form$ if
 - Q is a quantifier ($Q \in \{\exists, \forall\}$),
 - x is a variable ($x \in Var$) and
 - A is a formula ($A \in Form$).

Fixed Occurences

Definition

Let $L^{(1)} = \langle LC, Var, Con, Term, Form \rangle$ be a first order language, $A \in Form$ be a formula, and $x \in Var$ be a variable.

- A fixed occurrence of the variable x in the formula A is **free** if it is not in the subformulae $\forall xB$ or $\exists xB$ of the formula A .
- A fixed occurrence of the variable x in the formula A is **bound** if it is not free.

Free variables

Definition

Let $L^{(1)} = \langle LC, Var, Con, Term, Form \rangle$ be a first order language and let $A \in Form$ be a formula. The **set of free variables** of the formula A —in notation: $FreeVar(A)$ —is given by the following inductive definition:

- If A is an atomic formula—i.e. $A \in AtForm$ —then the members of the set $FreeVar(A)$ are the variables occurring in A .
- If the formula A is $\neg B$, then $FreeVar(A) = FreeVar(B)$.
- If the formula A is $(B \supset C)$, $(B \wedge C)$, $(B \vee C)$ or $(B \equiv C)$, then $FreeVar(A) = FreeVar(B) \cup FreeVar(C)$.
- If the formula A is $\forall xB$ or $\exists xB$, then $FreeVar(A) = FreeVar(B) \setminus \{x\}$.

Bound variables

Definition

Let $L^{(1)} = \langle LC, Var, Con, Term, Form \rangle$ be a first order language and let $A \in Form$ be a formula. The **set of bound variables** of the formula A —in notation: $BoundVar(A)$ —is given by the following inductive definition:

- If A is an atomic formula—i.e. $A \in AtForm$ —then $BoundVar(A) = \emptyset$.
- If the formula A is $\neg B$, then $BoundVar(A) = BoundVar(B)$.
- If the formula A is $(B \supset C)$, $(B \wedge C)$, $(B \vee C)$ or $(B \equiv C)$,
 $BoundVar(A) = BoundVar(B) \cup BoundVar(C)$.
- If the formula A is $\forall xB$ or $\exists xB$, then $BoundVar(A) = BoundVar(B) \cup \{x\}$.

Open and Closed Formulae

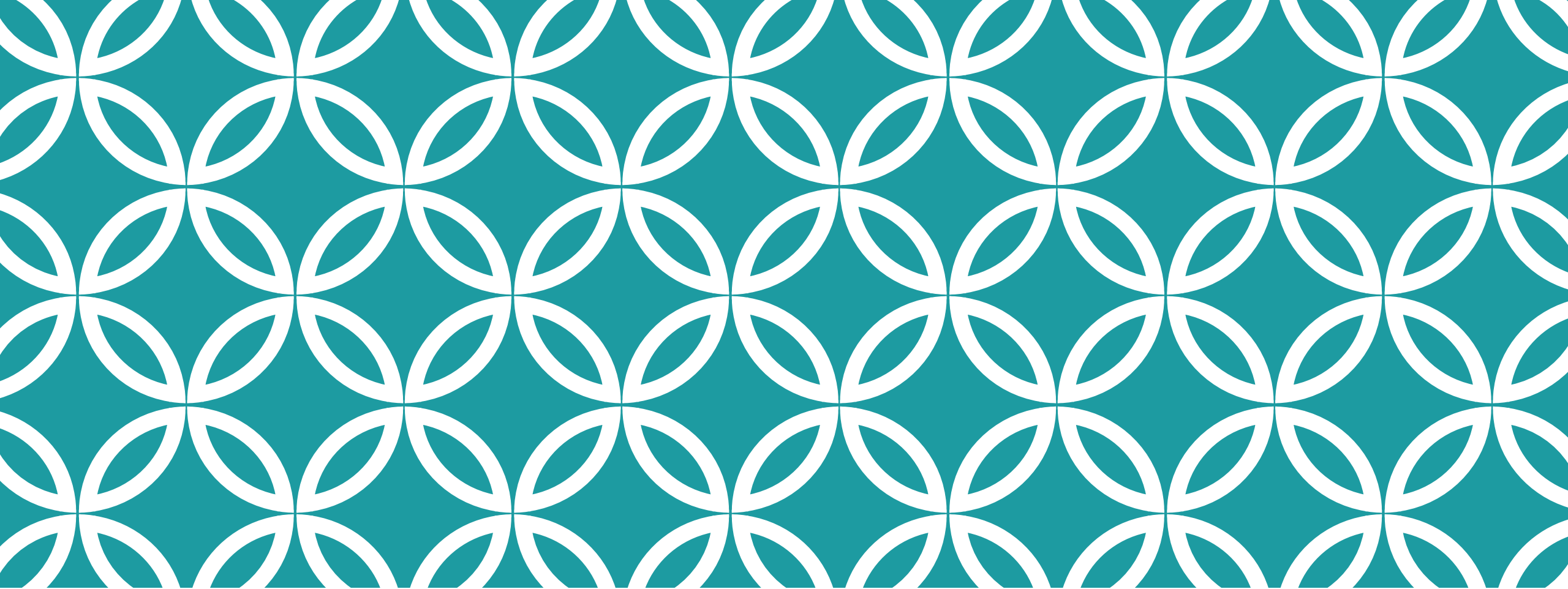
Definition

Let $L^{(1)} = \langle LC, Var, Con, Term, Form \rangle$ be a first order language and $A \in Form$ be a formula.

- If $FreeVar(A) \neq \emptyset$, then the formula A is an **open formula**.
- If $FreeVar(A) = \emptyset$, then the formula A is a **closed formula**.

Remark

- The formula A is open if there is at least one variable which has at least one free occurrence in A .
- The formula A is closed if there is no variable which has a free occurrence in A .



Exercises



Exercises

Mark the variables that are member of the set BoundVar of formula

$\forall v(\forall y(R(x, x) \wedge \exists u\forall u\exists v\forall u\forall yR(g(z, x), h(c, z))) \equiv c = x)$ where $F(0)=\{c\}$, $F1=\{f\}$, $F2=\{g,h\}$, $P1=\{P\}$ és $P2=\{Q,R\}$!

Select one or more:

- 
- ☐ a. v
 - ☐ b. x
 - ☐ c. c
 - ☐ d. z

Mark the variables that are member of the set BoundVar of formula

$\forall u \exists y \forall w \forall u \forall x \neg \exists u \exists w \exists w \neg f(z) = g(v, y)$ where $F(0)=\{c\}$, $F1=\{f\}$, $F2=\{g,h\}$, $P1=\{P\}$ és $P2=\{Q,R\}$!

Select one or more:

 ☐ a. w

 ☐ b. u

☐ c. c

 ☐ d. y

Mark the variables that are member of the set FreeVar of formula

$y = c \vee \forall u \forall y \exists u \neg (z = f(z) \supset Q(z, h(v, y)))$ where $F(0) = \{c\}$, $F1 = \{f\}$, $F2 = \{g, h\}$, $P1 = \{P\}$
és $P2 = \{Q, R\}$!

Select one or more:

☐ a. c

☐ b. u

☒ c. v

☒ d. y

Mark the variables that are member of the set BoundVar of formula

$\forall u(P(g(y, f(u))) \vee R(v, f(x))) \wedge (Q(w, y) \equiv z = h(v, v))$ where $F(0)=\{c\}$, $F1=\{f\}$,
 $F2=\{g,h\}$, $P1=\{P\}$ és $P2=\{Q,R\}$!

Select one or more:

☐ a. w

☐ b. v

 ☐ c. u

☐ d. x

Mark the variables that are member of the set FreeVar of formula

$\forall u((Q(f(u), w) \supset c = w) \wedge z = h(c, g(u, g(x, x))))$ where $F(0)=\{c\}$, $F1=\{f\}$, $F2=\{g,h\}$,
 $P1=\{P\}$ és $P2=\{Q,R\}$!

Select one or more:

☐ a. u

 ☐ b. z

☐ c. c

 ☐ d. w

Mark the variables that are member of the set BoundVar of formula

$\forall z(\exists x P(h(h(v, v), x)) \equiv \exists u \forall v R(x, f(h(c, u))))$ where $F(0)=\{c\}$, $F1=\{f\}$, $F2=\{g, h\}$,
 $P1=\{P\}$ és $P2=\{Q, R\}$!

Select one or more:

-  ☐ a. u
-  ☐ b. x
-  ☐ c. v
-  ☐ d. z

Mark the variables that are member of the set FreeVar of formula

$\forall u(z = v \vee \exists z(\neg \forall u \forall v z = g(h(u, u), z) \supset R(z, w)))$ where $F(0)=\{c\}$, $F1=\{f\}$, $F2=\{g,h\}$, $P1=\{P\}$
 $P2=\{Q,R\}$!

Select one or more:

☐ a. v

☐ b. c

☐ c. u

☐ d. w

Mark the variables that are member of the set FreeVar of formula

$\forall v \forall w (P(h(z, g(x, h(w, g(w, w)))))) \wedge \exists u \exists x c = g(z, f(g(z, v)))$) where $F(0)=\{c\}$,
 $F1=\{f\}$, $F2=\{g,h\}$, $P1=\{P\}$ és $P2=\{Q,R\}$!

Select one or more:

- 
- ☐ a. z
 - ☐ b. v
 - ☐ c. w
 - ☐ d. u

Exercises

Mark the bounded (indexed) occurrences in the formula

$\neg \forall v (Q(g(u, c), z_1) \wedge (Q(v, v_2) \equiv g(x, z_3) = f(w_4)))$, where $F(0)=\{c\}$, $F1=\{f\}$, $F2=\{g,h\}$,
 $P1=\{P\}$ and $P2=\{Q,R\}$?

Select one or more:

- ☐ a. 4
- ☐ b. 2
- ☐ c. 3
- ☐ d. 1

Mark the free (indexed) occurrences in the formula

$\exists u_4 \exists x (\exists z \forall x \forall u_2 Q(u, g(w, g(x, u_1))) \supset Q(v, z) \equiv Q(y_3, y))$, where $F(0)=\{c\}$, $F1=\{f\}$, $F2=\{g,h\}$, $P1=\{P\}$ and $P2=\{Q,R\}$?

Select one or more:

- ☐ a. 2
- ☐ b. 1
- ☐ c. 3
- ☐ d. 4

Mark the free (indexed) occurrences in the formula

$v = w_1 \supset \forall u \exists w (R(x_2, f(v)) \vee \exists u \forall x Q(u_3, f(h(w, v_4))))$, where $F(0)=\{c\}$, $F1=\{f\}$, $F2=\{g,h\}$, $P1=\{P\}$ and $P2=\{Q,R\}$?

Select one or more:

- ☐ a. 3
- ☐ b. 4
- ☐ c. 2
- ☐ d. 1

Mark the free (indexed) occurrences in the formula

$\exists u_4 \exists x (\exists z \forall x \forall u_2 Q(u, g(w, g(x, u_1))) \supset Q(v, z) \equiv Q(y_3, y))$, where $F(0)=\{c\}$, $F1=\{f\}$, $F2=\{g,h\}$, $P1=\{P\}$ and $P2=\{Q,R\}$?

Select one or more:

- ☐ a. 4
- ☐ b. 2
- ☐ c. 1
- ☐ d. 3

Mark the free (indexed) occurrences in the formula

$\forall y (w = g(v, z) \wedge \forall v_3 \forall w_2 P(f(h(g(z, c), w_1)))) \equiv R(v, u_4)$, where $F(0)=\{c\}$, $F1=\{f\}$, $F2=\{g,h\}$, $P1=\{P\}$ and $P2=\{Q,R\}$?

Select one or more:

- ☐ a. 2
- ☐ b. 4
- ☐ c. 1
- ☐ d. 3

Open and closed formulae

$\mathcal{F}(0) = \{c, d\}, \mathcal{F}(1) = \{f\}, \mathcal{F}(2) = \{g\},$
 $\mathcal{P}(1) = \{P\}, \mathcal{P}(2) = \{Q\}$ and $Var = \{x, \dots\}$

Which formulae are open?

$$\forall x P(x) \wedge Q(f(c), g(c, g(c, d))) \vee \exists x P(x)$$

$$\forall x (P(x) \wedge Q(f(c), g(c, g(c, d)))) \vee \exists x P(x))$$

$$Q(c, g(c, g(c, d))) \vee \exists x (Q(g(c, c), d) \wedge Q(d, g(c, c)))$$

$$\forall x (\exists y P(y) \wedge \forall z Q(x, z)) \vee P(z)$$

$$\exists x P(x) \vee \exists x Q(x) \supset \exists x (P(x) \vee R(x))$$

$$\forall x (P(x) \wedge R(x)) \supset \forall x P(x) \wedge \forall x Q(x)$$

$$\forall x (\exists y Q(x, y) \supset \forall z (Q(c, z) \wedge Q(c, z)))$$

$$\exists x (\forall x P(x) \supset Q(y, z) \wedge Q(y, x))$$

$$\exists z (P(y) \wedge \exists x Q(x, z) \supset Q(x, z))$$

$$P(f(c)) \vee Q(c, d) \equiv \forall x \neg P(g(g(x, d), g(c, x)))$$

Open and closed formulae

$\mathcal{F}(0) = \{c, d\}, \mathcal{F}(1) = \{f\}, \mathcal{F}(2) = \{g\},$
 $\mathcal{P}(1) = \{P\}, \mathcal{P}(2) = \{Q\}$ and $Var = \{x, \dots\}$

Which formulae are open?

$$\forall x P(x) \wedge Q(f(c), g(c, g(c, d))) \vee \exists x P(x)$$

$$\forall x (P(x) \wedge Q(f(c), g(c, g(c, d)))) \vee \exists x P(x))$$

$$Q(c, g(c, g(c, d))) \vee \exists x (Q(g(c, c), d) \wedge Q(d, g(c, c)))$$


$$\forall x (\exists y P(y) \wedge \forall z Q(x, z)) \vee P(z)$$

$$\exists x P(x) \vee \exists x Q(x) \supset \exists x (P(x) \vee R(x))$$

$$\forall x (P(x) \wedge R(x)) \supset \forall x P(x) \wedge \forall x Q(x)$$


$$\forall x (\exists y Q(x, y) \supset \forall z (Q(c, z) \wedge Q(c, z)))$$


$$\exists x (\forall x P(x) \supset Q(y, z) \wedge Q(y, x))$$


$$\exists z (P(y) \wedge \exists x Q(x, z) \supset Q(x, z))$$

$$P(f(c)) \vee Q(c, d) \equiv \forall x \neg P(g(g(x, d), g(c, x)))$$

Open and closed formulae

$\mathcal{F}(0) = \{c, d\}, \mathcal{F}(1) = \{f\}, \mathcal{F}(2) = \{g\},$
 $\mathcal{P}(1) = \{P\}, \mathcal{P}(2) = \{Q\}$ and $Var = \{x, \dots\}$

Which formulae are closed?

$$\forall x(\exists y P(y) \wedge \forall z Q(x, z)) \vee P(z)$$

$$\exists x P(x) \vee \exists x Q(x) \supset \exists x (P(x) \vee R(x))$$

$$\forall x (P(x) \wedge R(x)) \supset \forall x P(x) \wedge \forall x Q(x)$$

$$\exists z (P(y) \wedge \exists x Q(x, z) \supset Q(x, z))$$

$$\exists x (P(x) \wedge R(x)) \supset \exists x P(x) \wedge \exists x R(x)$$

$$\forall x \forall y Q(x, y) \supset \forall y \forall x Q(x, y)$$

$$\forall x (\exists y (P(y) \supset Q(x, y)) \wedge Q(y, z))$$

$$\forall x (\exists y Q(x, y) \supset \forall z (Q(c, z) \wedge Q(c, z)))$$

$$\exists x (\forall x P(x) \supset Q(y, z) \wedge Q(y, x))$$

$$\forall y (\exists x Q(y, x) \supset \forall y (Q(c, y) \wedge Q(y, x)))$$

$$P(f(c)) \vee Q(c, d) \equiv \forall x \neg P(g(g(x, d), g(c, x)))$$